Magnetosphere and changing fields

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use specific measurements or observations to compare a magnetic-field explanation with an unsupported statement about space weather.

SUCCESS CRITERIA - I can...

1. Connect changing magnetic fields with induced electric effects.
2. Distinguish a fact you treat as true from something you observed.
3. Name the measurements or observations needed to predict space weather.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Near an ideal long straight wire, current becomes **3** times as large and distance becomes **5** times as large. **Compare B**.

Marzano check - circle: ① ② ③

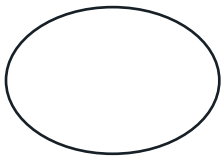
02 Worked example: explain the prediction

Reference: $|\text{emf}_{avg}| = N|\Delta\Phi|/\Delta t$; $\Phi = BA$.
Uniform perpendicular field for $\Phi = BA$. Use supplied/measured field changes; an asteroid alone is not a sign of an electromagnetic pulse.

Worked reasoning: A changing magnetic environment can induce voltages in conducting loops. A static magnetic field alone is not the same as a changing flux through a particular circuit.

Your turn: Sketch a conducting loop and field arrows; **show one way flux can** change.

Label field direction, loop area and what changes.



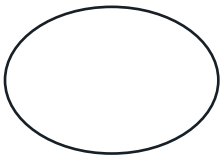
Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

1. **Draw a coil** in a changing magnetic field.
2. **Label what changes** and where an induced emf could appear.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

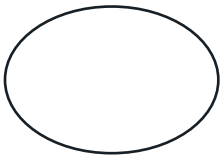
Marzano check - circle: ① ② ③

04 Apply the physics

A hypothetical **10**-turn loop of area **5 m²** experiences a uniform perpendicular field change of **2 × 10⁻⁶ T** over **7 s**.

1. **Label the given quantities** and **calculate average emf magnitude**.
2. Does this calculation establish what caused the field change?

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: ① ② ③

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Continue down each column.

05 Explain your reasoning

A headline says an approaching rock will disable power grids through an EMP. It gives no field measurements or explanation of how this would happen. **Name two measurements or observations** needed before accepting this statement.

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Two series components have drops **4 V** and **6 V**. A charge of **8 C** passes through each. **Find each energy transfer** and the total.
Label voltage, charge and energy with their distinct units.

Initial / supply _____ V: ____ J/C

Final / component _____ V: ____ J/C

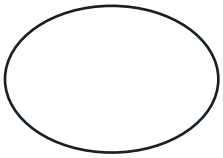
Charge: ____ C Energy transfer: ____ J

Marzano check - circle: 1 2 3

07 Exit check

Explain why a one-loop calculation cannot by itself predict effects on a whole power grid.

Label field direction, loop area and what changes.



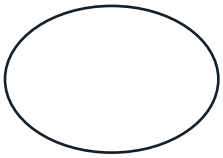
Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A one-turn **100 m²** loop is perpendicular to a uniform field that changes by **2** microtesla in **0.5 s**.
1. **Estimate emf**.
2. **Explain why this toy result** is not a full forecast for a power grid.

Label field direction, loop area and what changes.



Turns: ____ Time interval: ____ emf: ____

Marzano check - circle: 1 2 3